

MELANOMA CANCER PREDICTION

Early diagnosis of melanoma skin cancer is paramount for effective treatment and increased survival rates. Melanoma, the most dangerous form of skin cancer, arises from the uncontrolled growth of melanocytes, the cells responsible for producing melanin, the pigment that gives skin its color. While it comprises only a small percentage of skin cancer cases, melanoma accounts for a disproportionate number of skin cancer-related deaths due to its aggressive nature and potential to metastasize to other organs. Melanoma of skin is the 17th most common cancer worldwide. It is the 13th most common cancer in men and the 15th most common cancer in women.

One of the primary reasons early diagnosis is crucial is because melanoma can often be successfully treated if detected in its early stages. When melanoma is confined to the outermost layer of the skin (the epidermis) and has not yet invaded deeper layers or spread to other parts of the body, it is highly curable through surgical removal. However, if left untreated and allowed to advance, melanoma can penetrate into the deeper layers of the skin and eventually metastasize to distant organs, such as the lungs, liver, or brain, making treatment significantly more challenging and decreasing the chances of survival.

Beyond the potential for successful treatment, early diagnosis of melanoma can also lead to improved quality of life for patients. Detecting melanoma before it has spread allows for less invasive treatment options and reduces the likelihood of disfigurement or disability resulting from extensive surgical procedures or aggressive therapies. Additionally, early detection may spare patients from the physical and emotional toll of battling advanced-stage cancer and undergoing intensive treatments with significant side effects.

Regular skin examinations, both self-examinations and those conducted by healthcare professionals, play a crucial role in early detection. Individuals should be vigilant in monitoring their skin for any changes in moles, freckles, or other pigmented areas, paying particular attention to the ABCDEs of melanoma: Asymmetry, Border irregularity, Color variation, Diameter greater than 6 millimeters, and Evolution or change over time. Suspicious lesions should be promptly evaluated by a dermatologist, who can perform a thorough examination and, if necessary, order additional tests such as a biopsy to confirm a diagnosis. Advancements in imaging technologies and diagnostic techniques have also enhanced the ability to detect melanoma in its early stages. In our project, we examined a dataset containing more than 10000 images of benign and malignant tumors, then used the modified EfficientNetV2 models to classify whether the image shows symptoms of melanoma cancer.